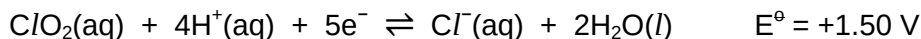


Answer any **four** questions

- 1 (a) The yellow gas chlorine dioxide,  $\text{ClO}_2$ , has been used for many years as a flour-improving agent in bread-making.

It can act as a powerful oxidising agent in acidic medium.



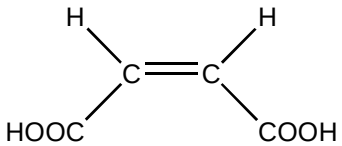
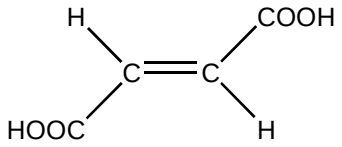
- (i) Define the term *standard electrode potential*.
- (ii) Draw a fully labelled diagram of the electrochemical cell you would set up in order to measure the standard reduction potential of  $\text{ClO}_2/\text{Cl}^-$  in the laboratory. Indicate clearly the positive and negative electrodes.
- (iii) When chlorine dioxide is bubbled through a solution of hydrogen sulfide,  $\text{H}_2\text{S}$ , a yellow solid is observed.

Construct a balanced equation, with state symbols, for the reaction that occurs.

[6]

- (b) Maleic acid and fumaric acid with the molecular formula  $\text{C}_4\text{H}_4\text{O}_4$  are two isomeric alkenedioic acids.

The following table compares the  $\text{p}K_\text{a}$  values of the two acids.

| acid    | structure                                                                           | $\text{p}K_1$ | $\text{p}K_2$ |
|---------|-------------------------------------------------------------------------------------|---------------|---------------|
| maleic  |  | 1.90          | 6.07          |
| fumaric |  | 3.03          | 4.44          |

- (i) Suggest a reason why the  $\text{p}K_1$  value of fumaric acid is lower than its  $\text{p}K_2$ .
- (ii) Suggest a reason why the  $\text{p}K_1$  value of maleic acid is lower than the  $\text{p}K_1$  of fumaric acid.
- (iii) Calculate the pH of a  $0.15 \text{ mol dm}^{-3}$  solution of fumaric acid (ignore the effect of  $\text{p}K_2$  on the pH).
- (iv)  $40 \text{ cm}^3$  of  $0.10 \text{ mol dm}^{-3}$  NaOH is added to  $10 \text{ cm}^3$  of  $0.15 \text{ mol dm}^{-3}$  fumaric acid. Sketch a graph to illustrate how the pH would vary with volume of NaOH added.

Show how you have obtained the various key points on the graph.

[7]

(c) Two isomers **A** and **B** have elemental composition by mass of C: 79.40%, H: 8.82%, and O: 11.78%.

(i) Calculate their empirical formula.

(ii) Isomer **A** gives a yellow precipitate with warm, alkaline iodine solution. It reacts with concentrated sulfuric acid at 170 °C to give three products, **C**, **D** and **E**.

**E** gives 2 moles of carbon dioxide with hot, acidified potassium manganate(VII) solution.

Isomer **B** gives a white precipitate, **F**, when mixed with aqueous bromine in the ratio of 1 : 3. **B** gives 2 moles of carbon dioxide when reacted with hot, acidified potassium manganate(VII).

Draw the structures of **A** to **F**.

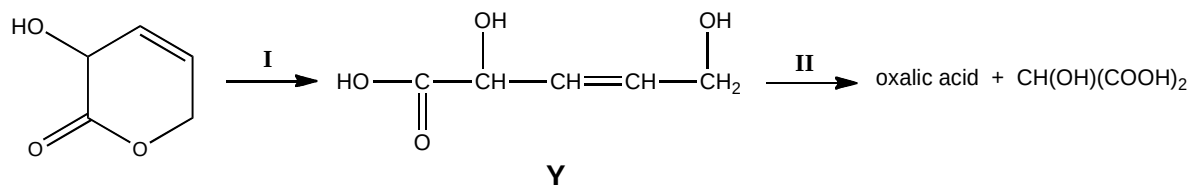
[7]

[Total: 20]

[Turn over

- 2 Oxalic acid (ethane-1,2-dioic acid,  $\text{C}_2\text{H}_2\text{O}_4$ ) is commonly found in vegetables such as spinach and carrots.

(a) The following shows a reaction pathway in which oxalic acid is formed, via compound **Y**.



- (i) State the reagents and conditions for reaction **I**.
- (ii) A student tried to synthesise oxalic acid in reaction **II** by using hot, acidified potassium manganate(VII). Explain why he would not be successful.
- (iii) Compound **Y** exhibits stereoisomerism. State the number of stereoisomers of compound **Y** and draw them.

[5]

- (b) When oxalic acid is treated with hot concentrated sulfuric acid, two gases are produced.

- (i) Suggest the identities of the two gases.
- (ii) Under what conditions would the two gases behave ideally? Explain your answer.

[3]

In 1870 during the Franco-Prussian War, the French emperor Napoleon III was captured in the battle of Sedan, which saw the defeat of the French. During this time, Napoleon III was debilitated with constant pain due to the formation of kidney stones inside his body. It was debatable whether his health condition had contributed significantly to his loss in the battle. Calcium oxalate ( $\text{CaC}_2\text{O}_4$ ) is the primary constituent of the common kidney stones, and is believed to form due to inappropriate diets.

- (c) (i) Draw a 'dot-and-cross' diagram to show the bonding in calcium oxalate.
- (ii) It was experimentally determined that all the C–O bonds in  $\text{C}_2\text{O}_4^{2-}$  are of the same length. Suggest a reason for this observation.

[3]

(d) 50.0 cm<sup>3</sup> of 0.100 mol dm<sup>-3</sup> CaCl<sub>2</sub> and 100.0 cm<sup>3</sup> of 0.300 mol dm<sup>-3</sup> Na<sub>2</sub>C<sub>2</sub>O<sub>4</sub> are mixed together at 25 °C.

- (i) Write an expression for the solubility product of calcium oxalate and state its units.
- (ii) Calculate the amount of calcium oxalate that will be produced.
- (iii) Calculate the concentration of C<sub>2</sub>O<sub>4</sub><sup>2-</sup>, in mol dm<sup>-3</sup>, left over in the resulting solution once equilibrium has been established.
- (iv) Given that the value of the solubility product of calcium oxalate is 2.32 × 10<sup>-9</sup> at 25 °C, calculate the concentration of Ca<sup>2+</sup>, in mol dm<sup>-3</sup>, in the resultant solution.
- (v) Given that the solubility product of magnesium oxalate is higher than that of calcium oxalate, discuss qualitatively whether it is possible to extract 100% of Ca<sup>2+</sup> by adding 100 cm<sup>3</sup> of 0.300 mol dm<sup>-3</sup> Na<sub>2</sub>C<sub>2</sub>O<sub>4</sub> drop-wise to a 50.0 cm<sup>3</sup> solution containing 0.200 mol dm<sup>-3</sup> of MgCl<sub>2</sub> and 0.100 mol dm<sup>-3</sup> of CaCl<sub>2</sub>.

[6]

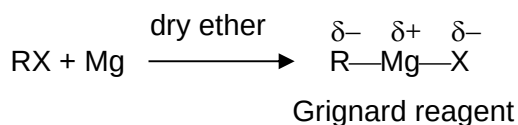
(e) Calcium oxalate and magnesium oxalate decompose at different temperatures to produce the corresponding metal oxides. Explain this difference in their decomposition temperatures.

[3]

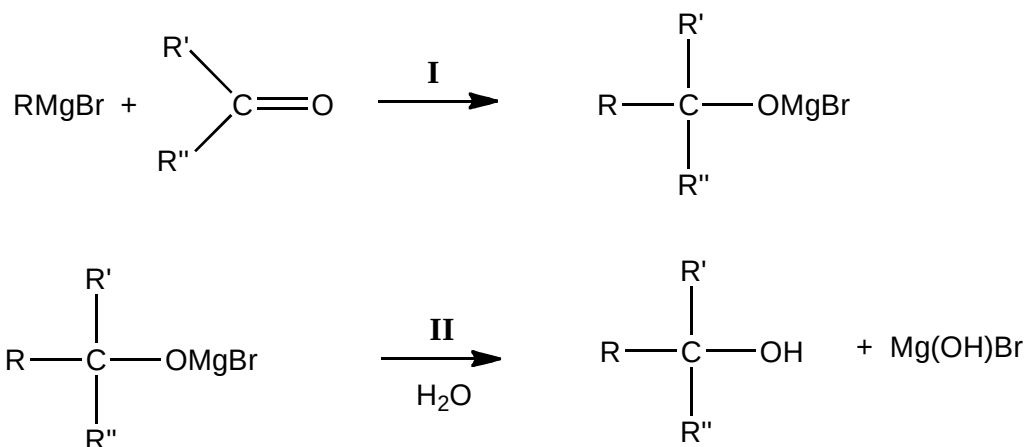
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[Turn over

- 3 (a) Halogenoalkanes,  $RX$ , are important intermediates used to produce many organic compounds. They may be converted into Grignard reagents by reacting them with magnesium in dry ether:

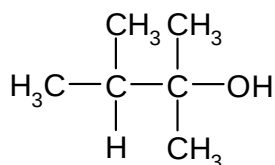


On reaction with carbonyl compounds,  $RMgX$  forms various classes of alcohols:



(where  $R'$  and  $R''$  are either H or alkyl groups.)

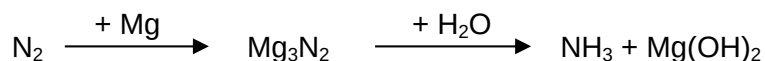
- (i) What *types of reaction* are steps **I** and **II**?
- (ii) With reference to the above reactions, and using propan-2-ol as the **only** organic compound, outline a synthesis of 2,3-dimethylbutan-2-ol. Give the reagents and conditions, as well as the structures of any intermediate products.



2,3-dimethylbutan-2-ol

[6]

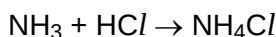
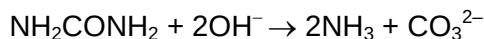
- (b) Magnesium reacts with pure nitrogen to form  $Mg_3N_2$  which can be considered as a possible intermediate in the 'fixing' of nitrogen to make ammonia-based fertilisers.



Nitrogen is an essential macronutrient needed by all plants to thrive. It is an important component of many structural, genetic and metabolic compounds in plant cells. It is also one of the basic components of chlorophyll, the compound by which plants use sunlight energy to produce sugars during the process of photosynthesis.

Another type of ammonia-based fertiliser contains urea,  $\text{NH}_2\text{CONH}_2$ , and has a large percentage by mass of nitrogen.

The nitrogen content of an urea-containing fertiliser can be measured by boiling a sample of known mass of the fertiliser with an excess of  $\text{NaOH(aq)}$ , absorbing the gas evolved in water, and titrating the aqueous solution with hydrochloric acid of a known concentration.



When 0.100 g of this fertiliser was subjected to this procedure, the resulting solution of ammonia required  $15.0 \text{ cm}^3$  of  $0.200 \text{ mol dm}^{-3} \text{ HCl}$  for neutralisation.

- (i) Calculate the percentage of nitrogen in the urea-containing fertiliser.
- (ii) Urea can hydrolyse in water to form ammonia and another colourless gas. Write an equation for this reaction.
- (iii) Urea can be applied directly to the soil either before or during the growing of crops. Based on the information given, what would be the major **disadvantage** of using magnesium nitride in the same way?

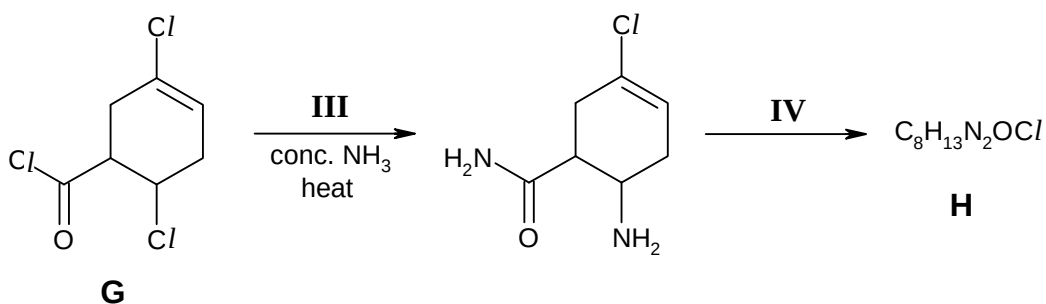
[5]

(c) Like water, liquid ammonia can also undergo self-ionisation.

- (i) Write an equation to represent the self-ionisation of liquid ammonia.
- (ii) Draw the dot-cross diagrams to show the bonding in the species formed in (c)(i). Suggest the bond angle(s) in the species.

[5]

(d) Ammonia is used in the synthesis of **H** from **G** as shown below.



- (i) Give the reagents and conditions for step **IV** and hence suggest the structure of **H**.
- (ii) One of the chlorine atoms in **G** is inert towards hot concentrated  $\text{NH}_3$ , as shown above. Explain why this is so.

[4]

[Total: 20]

- 4 The chemistry of iron has been extensively studied and its applications span from industrial processes to biological functions.

(a) Iron is extracted from its naturally occurring ore, known as hematite,  $\text{Fe}_2\text{O}_3$ .

- (i) Using suitable data from the *Data Booklet* as well as the information given below, draw a labelled Born-Haber energy cycle. Hence determine the lattice energy of  $\text{Fe}_2\text{O}_3$ .

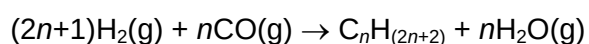
| enthalpy change                                                 | $\text{kJ mol}^{-1}$ |
|-----------------------------------------------------------------|----------------------|
| atomisation of iron                                             | +414                 |
| 1 <sup>st</sup> and 2 <sup>nd</sup> electron affinity of oxygen | +703                 |
| formation of $\text{Fe}_2\text{O}_3$                            | -826                 |

- (ii) The melting points of  $\text{Fe}_2\text{O}_3$  and  $\text{Al}_2\text{O}_3$  are found to be  $1566^\circ\text{C}$  and  $2072^\circ\text{C}$  respectively. Using suitable data from the *Data Booklet*, explain the difference in their melting points.
- (iii)  $\text{Fe}_2\text{O}_3$  and  $\text{Al}_2\text{O}_3$  readily dissolve in acids, however only  $\text{Al}_2\text{O}_3$  is soluble in alkalis. Write equations to explain the above observations.

[8]

- (b) Solid iron is also extensively used as a catalyst in many industrial applications such as the Haber process and Fischer-Tropsch process. For the latter, carbon monoxide is converted into hydrocarbons for fuels and lubricants.

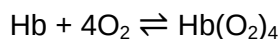
The equation for the Fischer-Tropsch process is given as



- (i) What property of iron renders it as a suitable catalyst in this process?
- (ii) Describe how iron catalyses the Fischer-Tropsch process in the production of butane. You may illustrate your answers with suitable diagrams.
- (iii) As the carbon chain length of the alkane increases, the value of the entropy change,  $\Delta S$ , for the reaction becomes more negative. Suggest why this is so.

[5]

- (c) Iron is important to our human bodies as it is required to form the oxygen-carrying protein, haemoglobin, found in our red blood cells.



- (i) Describe the **quaternary** structure of haemoglobin, with reference to its structure and bonding.
- (ii) With reference to the equation given above, explain how oxygenated blood is transported to the vital organs in our human bodies.
- (iii) Inhalation of carbon monoxide at high concentrations can lead to convulsions and death. Briefly explain the chemistry behind carbon monoxide poisoning, with reference to haemoglobin.
- (iv) In a strongly basic medium, haemoglobin undergoes *denaturation*. Explain the term in italics and suggest why this process occurs.

[7]

[Total: 20]

[Turn over



- 5 Copper is a typical transition element which can exhibit more than one oxidation state in its compounds. The common oxidation states of copper are +1 and +2, although it is known that basic anions like the oxides and fluorides stabilises the +3 and +4 oxidation states.

(a) (i) What do you understand by the term *transition element*?

(ii) Explain why transition element complexes are coloured.

[4]

(b) (i) Describe what you would observe when excess concentrated hydrochloric acid is added to aqueous copper(II) sulfate. Write an equation to explain the observation.

(ii) State and explain what would happen if excess aqueous ammonia is added to the resultant solution in (b)(i).

[4]

(c) The use of the *Data Booklet* is relevant to this part of the question.

(i) Explain why, when a sample of copper(I) sulfate was added to water, a pink solid and a pale blue solution were obtained.

(ii) The mixture in (c)(i) was filtered. The filtrate gave a white precipitate in a brown solution on addition of potassium iodide.  
Explain why this reaction would **not** be expected to occur.

(iii) Suggest a reason why the reaction in (c)(ii) does in fact occur.

[3]

- (d) When fluorine gas is passed over a heated mixture of potassium chloride and copper(II) chloride, a pale green solid, **Z**, containing only potassium, copper and fluorine is formed.

When 1.00 g of **Z** was reacted with water, 20.4 cm<sup>3</sup> of oxygen (measured at 298 K and 1 atm) were evolved and a blue acidic solution was formed. This solution was divided into two equal parts. Titration of one part with 0.100 mol dm<sup>-3</sup> NaOH required 17.00 cm<sup>3</sup> for neutralisation. The other part was electrolysed and 0.108 g of copper metal was deposited at the cathode.

- (i) Calculate the number of moles of oxygen molecules, hydrogen ions and copper ions formed in the reaction with water.
- (ii) Assuming that 1 mol of **Z** contains 1 mol of copper atoms, deduce its relative formula mass and hence its formula.
- (iii) Write balanced equations for the formation of **Z** and for its reaction with water.

[7]

- (e) An alkaline solution of complexed Cu<sup>2+</sup>(aq) is used in organic chemistry to test for a particular functional group.

Draw the displayed formula of **one** compound with the molecular formula C<sub>4</sub>H<sub>8</sub>O that would show a positive result in this test, and the displayed formula of **one** compound with the same molecular formula that would not. Label your structures clearly.

[2]

[Total: 20]

~ END OF PAPER ~